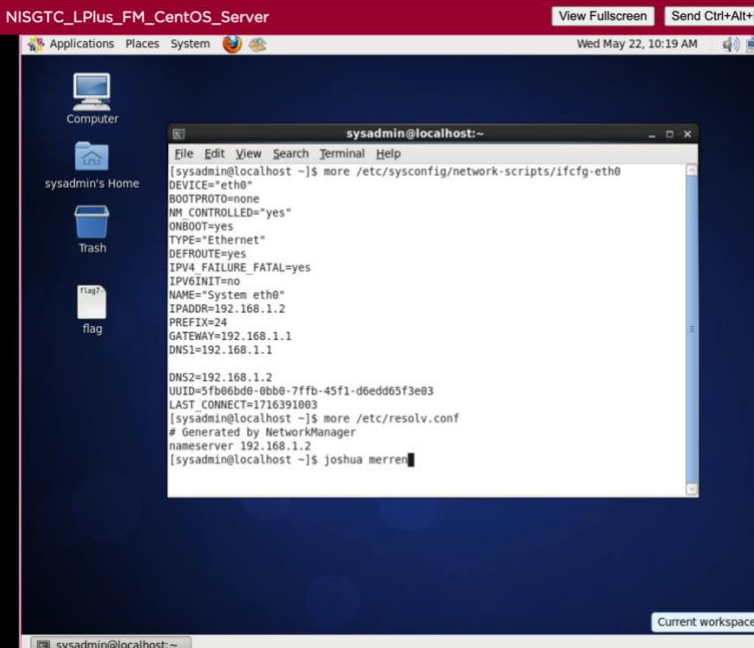
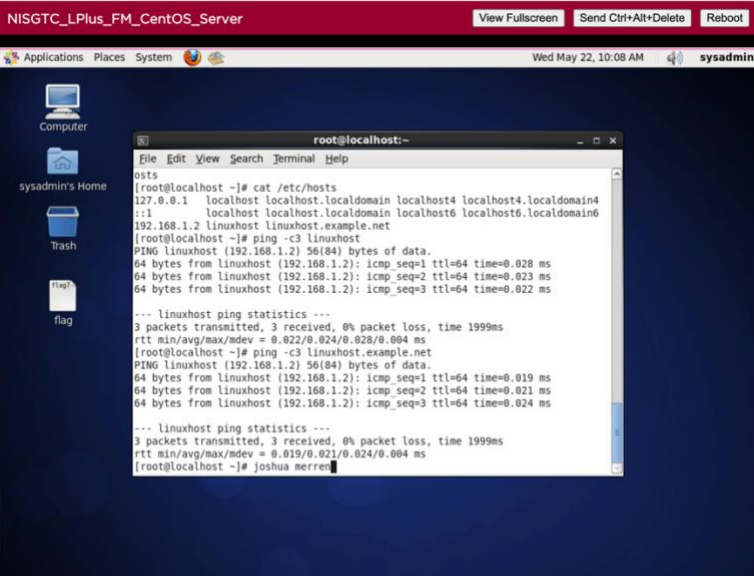


## CYB 230 Module Three Lab Worksheet

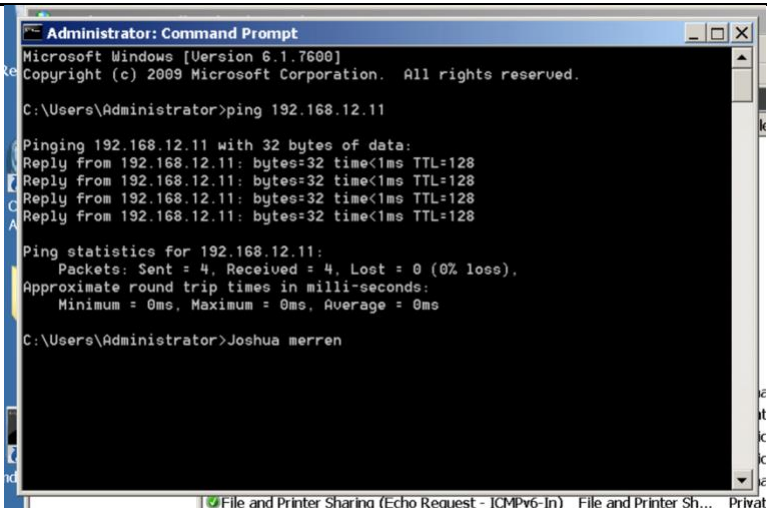
Complete this worksheet by replacing the bracketed phrases in the Response column with the relevant information.

## Lab: Basic Network Configuration

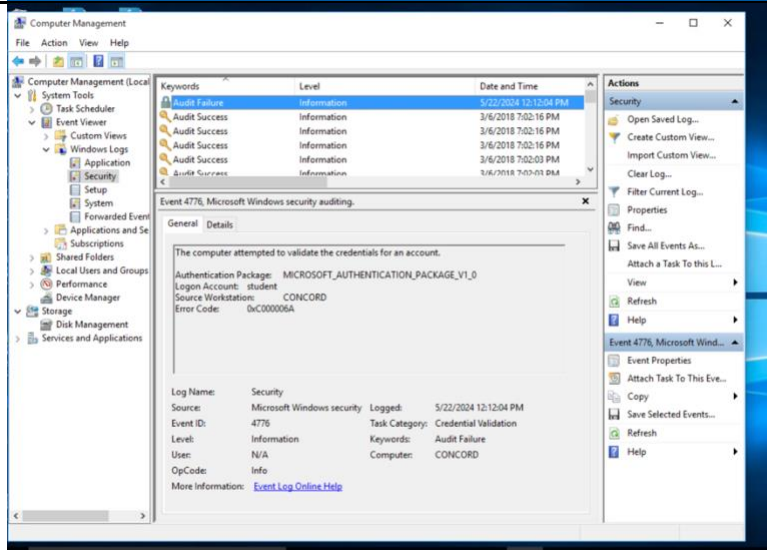
| Prompt                                                                                                                                                                                                                                     | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>In the lab section “Configuring a Network Interface Manually With the Networkmanager Service,” insert your name at the command line below the final configuration file output and include it in your screenshot.</p>                    |  <pre> NISGTC_LPlus_FM_CentOS_Server View Fullscreen Send Ctrl+Alt+Delete Applications Places System Wed May 22, 10:19 AM  sysadmin@localhost:~\$ nmcli con show NAME: eth0 DEVICE: eth0 BOOTPROTO: none HWADDR: 82:00:20:08:00:08 ONBOOT: yes TYPE: Ethernet DEFROUTE: yes IPV4_FAILURE_FATAL: yes IPV6INIT: no NAME: "System eth0" IPADDR: 192.168.1.2 PREFIX: 24 GATEWAY: 192.168.1.1 DNS1: 192.168.1.1 DNS2: 192.168.1.2 UUID: 5f8b6dd0-0bb0-7ffb-45f1-d6edd65f3e03 LAST_CONNECT: 1716391003 [sysadmin@localhost ~]\$ more /etc/resolv.conf # Generated by NetworkManager nameserver 192.168.1.2 [sysadmin@localhost ~]\$ joshua merrett </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p>In the lab section “Configuring a Centos Network Interface Manually With the Network Service,” insert your name at the command line below the output of the updated interface configuration file and include it in your screenshot.</p> |  <pre> NISGTC_LPlus_FM_CentOS_Server View Fullscreen Send Ctrl+Alt+Delete Reboot Applications Places System Wed May 22, 10:08 AM sysadmin  root@localhost:~\$ cat /etc/hosts 127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4 ::1 localhost localhost.localdomain localhost6 localhost6.localdomain6 192.168.1.2 linuxhost linuxhost.example.net PING linuxhost (192.168.1.2) 56(84) bytes of data: 64 bytes from linuxhost (192.168.1.2): icmp_seq=1 ttl=64 time=0.028 ms 64 bytes from linuxhost (192.168.1.2): icmp_seq=2 ttl=64 time=0.023 ms 64 bytes from linuxhost (192.168.1.2): icmp_seq=3 ttl=64 time=0.022 ms  --- linuxhost ping statistics --- 3 packets transmitted, 3 received, 0% packet loss, time 1999ms rtt min/avg/max/mdev = 0.022/0.024/0.028/0.004 ms root@localhost ~# ping -c 3 linuxhost.example.net PING linuxhost (192.168.1.2) 56(84) bytes of data: 64 bytes from linuxhost (192.168.1.2): icmp_seq=1 ttl=64 time=0.019 ms 64 bytes from linuxhost (192.168.1.2): icmp_seq=2 ttl=64 time=0.021 ms 64 bytes from linuxhost (192.168.1.2): icmp_seq=3 ttl=64 time=0.024 ms  --- linuxhost ping statistics --- 3 packets transmitted, 3 received, 0% packet loss, time 1999ms rtt min/avg/max/mdev = 0.019/0.021/0.024/0.004 ms root@localhost ~# joshua merrett </pre> |

| Prompt                                                                                                                                                                                                                                         | Response                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In “Configuring a Centos Network Interface Manually With the Network Service,” you are asked to resolve the addresses for three different names that all result in the same IP address. Which version would you find most useful? Explain why. | The hostname that is easiest to remember and indicates the server's function or location is most beneficial. For example, if the server is a database server in New York, a hostname like `ny-database01` would be practical. This makes remembering and quickly understanding the server's purpose easier without looking up its IP address. |

### Lab: Network Security—Firewalls

| Prompt                                                                                                                                                                                                       | Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In the lab section “Configure Windows Firewall With Advanced Security Using Administrative Tools,” insert your name at the command line below the ending output and include it in your screenshot.           |  <pre> Administrator: Command Prompt Microsoft Windows [Version 6.1.7600] Copyright (c) 2009 Microsoft Corporation. All rights reserved.  C:\Users\Administrator&gt;ping 192.168.12.11  Pinging 192.168.12.11 with 32 bytes of data: Reply from 192.168.12.11: bytes=32 time&lt;1ms TTL=128 Reply from 192.168.12.11: bytes=32 time&lt;1ms TTL=128 Reply from 192.168.12.11: bytes=32 time&lt;1ms TTL=128 Reply from 192.168.12.11: bytes=32 time&lt;1ms TTL=128  Ping statistics for 192.168.12.11:     Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),     Approximate round trip times in milli-seconds:         Minimum = 0ms, Maximum = 0ms, Average = 0ms  C:\Users\Administrator&gt;Joshua merren </pre> |
| Explain why it is not necessary to create an inbound rule on the internal 192.168.12.10 Windows server so that it can receive the response (ICMP echo reply) from the internal 192.168.12.11 Windows server. | I do not believe creating an inbound rule for ICMP echo replies is necessary because the firewall automatically allows responses to traffic initiated by the host. Since the echo request originates from 192.168.12.10, the firewall permits the reply as part of the established communication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Explain the advantages and disadvantages of having the firewall disabled at startup in the Linux operating system.                                                                                           | The advantage of having the firewall disabled at startup is that it simplifies server configuration and troubleshooting, especially during initial setup or when running applications that communicate on multiple ports. The disadvantage is a considerable increase in security risks as the server is exposed to potential threats, making it vulnerable to attacks until the firewall is manually enabled.                                                                                                                                                                                                                                                                                                                                                                                          |

### Lab: Implementing Security Policies on Windows and Linux

| Prompt                                                                                                                                                  | Response                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>In the lab section “Securing the Windows Logon Process,” modify “Warning!” to “Warning - [YOUR NAME]”. Provide a screenshot of the final output.</p> |                                                                                                                                                                                                                                                                                                                    |
| <p>In the lab section “Auditing Logon Failures,” provide a screenshot of the ending Audit Failure output.</p>                                           |                                                                                                                                                                                                                                                                                                                   |
| <p>What is the importance of automating system checks and log file creation for server management?</p>                                                  | <p>Automating system checks and log file creation is crucial for maintaining server reliability and security. It ensures that potential issues are detected early, and logs provide a trail of evidence in case of security breaches or failures. Automation helps manage large environments efficiently, reduces the risk of human error, and allows IT staff to focus on more strategic tasks.</p> |